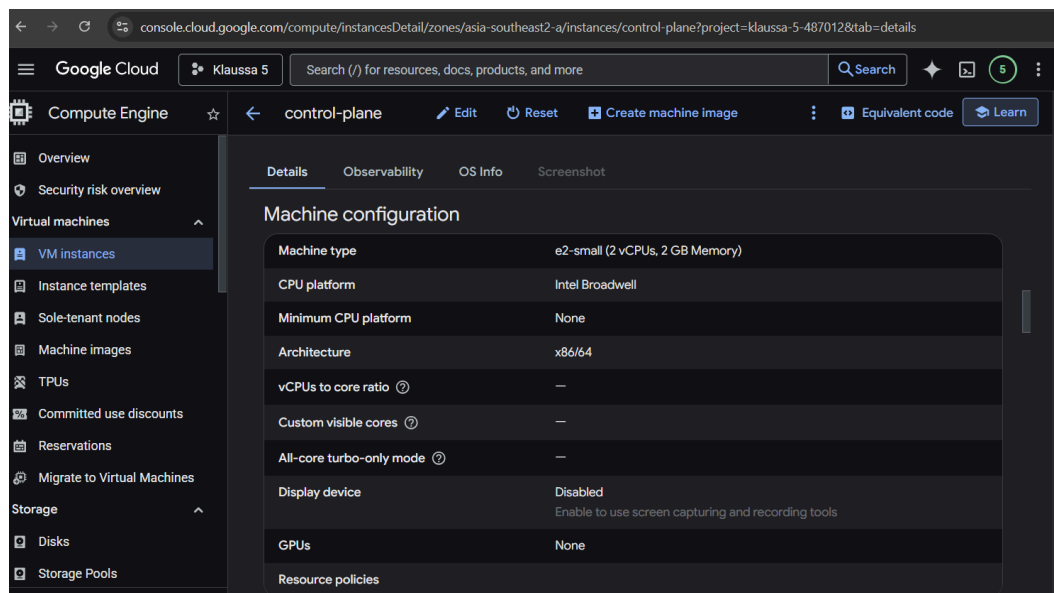
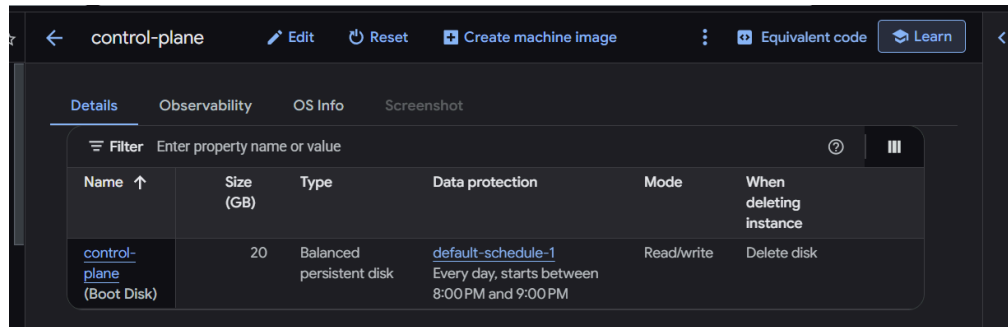


BRYAN JESHUA MARIO TIMUNG - 2206027021

1. Berikan tangkapan layar dalam melakukan pembuatan cluster sampai men-deploy aplikasi.

CONTROL PLANE



```
ssh.cloud.google.com/v2/ssh/projects/klaussa-5-487012/zones/asia-southeast2-a/instances/control-plane?auth...
ssh.cloud.google.com/v2/ssh/projects/klaussa-5-487012/zones/asia-southeast2-a/instances/control-plane?a...
SSH-in-browser
[UPLOAD FILE] [DOWNLOAD FILE] [!] [⌨] [⚙]

Welcome to Ubuntu 22.04.5 LTS (GNU/Linux 6.8.0-1048-gcp x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

This system has been minimized by removing packages and content that are
not required on a system that users do not log into.

To restore this content, you can run the 'unminimize' command.

Expanded Security Maintenance for Applications is not enabled.

12 updates can be applied immediately.
12 of these updates are standard security updates.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

navigoinfo_id@control-plane:~$ lsb_release -a
No LSB modules are available.
Distributor ID: Ubuntu
Description:    Ubuntu 22.04.5 LTS
Release:        22.04
Codename:       jammy
navigoinfo_id@control-plane:~$
```

WORKER 1

```
ssh.cloud.google.com/v2/ssh/projects/klaussa-5-487012/zones/asia-southeast2-a/instances/worker-1?authuser...
ssh.cloud.google.com/v2/ssh/projects/klaussa-5-487012/zones/asia-southeast2-a/instances/worker-1?authus...
SSH-in-browser
[UPLOAD FILE] [DOWNLOAD FILE] [!] [⌨] [⚙]

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

navigoinfo_id@worker-1:~$ lsb_release -a
No LSB modules are available.
Distributor ID: Ubuntu
Description:    Ubuntu 22.04.5 LTS
Release:        22.04
Codename:       jammy
navigoinfo_id@worker-1:~$
```

worker-1
Edit
Reset
Create machine image
Equivalent code

Details
Observability
OS Info
Screenshot

Reservation affinity ? Automatically choose
Consumed reservation ? —

Machine configuration

Machine type e2-custom-medium-2048 (2 vCPUs, 2 GB Memory)
CPU platform Intel Broadwell
Minimum CPU platform None
Architecture x86_64
vCPUs to core ratio ? —

worker-1
Edit
Reset
Create machine image
Equivalent code

Details
Observability
OS Info
Screenshot

Storage

Filter Enter property name or value ?

Name ↑	Size (GB)	Type	Data protection	Mode	When deleting instance
worker-1 (Boot Disk)	10	Balanced persistent disk	default-schedule-1 Every day, starts between 8:00 PM and 9:00 PM	Read/write	Delete disk

Backup plan ?
Managed by Backup and DR Service .
CREATE ON-DEMAND BACKUP

Backup schedule state ? Not configured
Backup plan —

WORKER 2

ssh.cloud.google.com/v2/ssh/projects/klaussa-5-487012/zones/asia-southeast2-a/instances/worker-2?authuser...

ssh.cloud.google.com/v2/ssh/projects/klaussa-5-487012/zones/asia-southeast2-a/instances/worker-2?authus...

SSH-in-browser

UPLOAD FILE
DOWNLOAD FILE
!

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

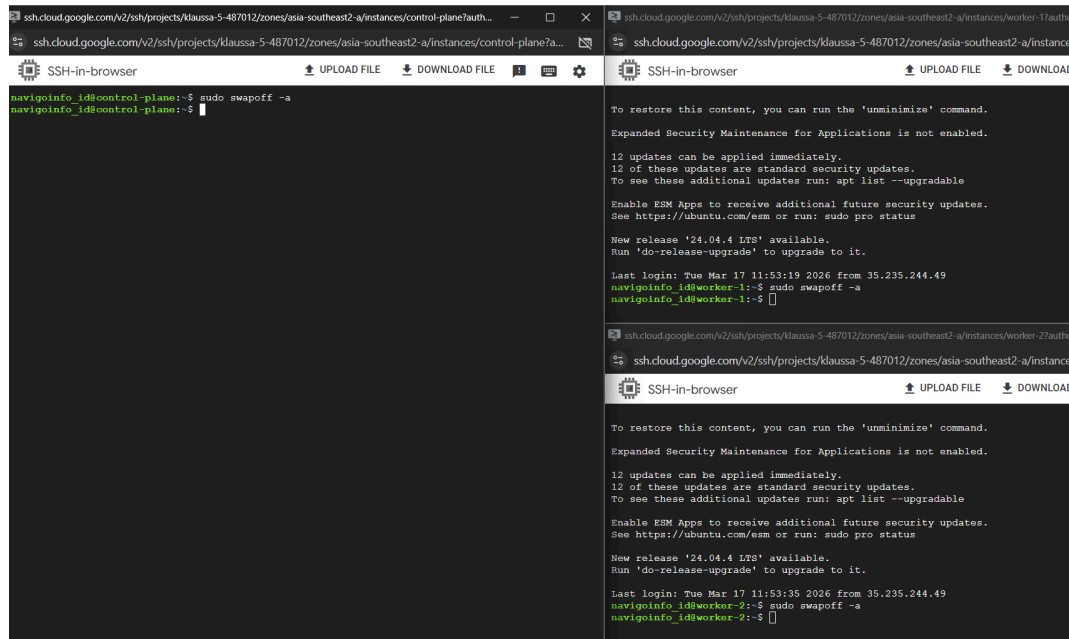
The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

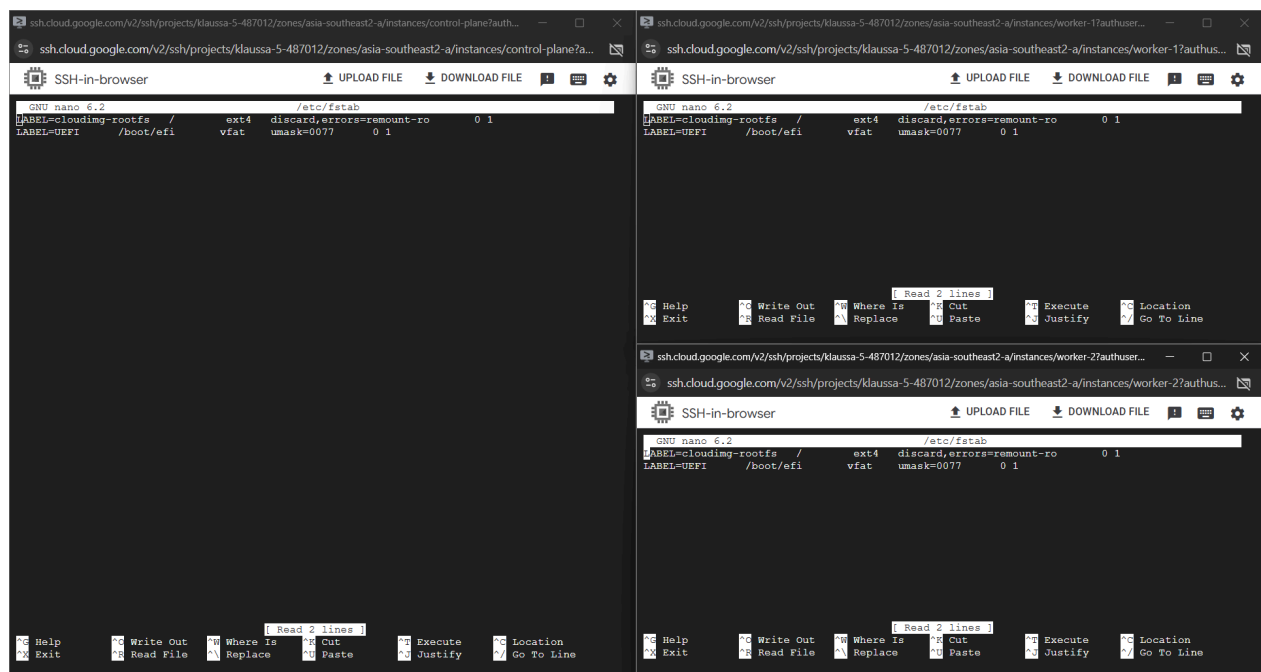
```

navigoinfo id@worker-2:~$ lsb_release -a
No LSB modules are available.
Distributor ID: Ubuntu
Description:    Ubuntu 22.04.5 LTS
Release:        22.04
Codename:       jammy
navigoinfo id@worker-2:~$

```

sudo nano /etc/fstab



```
cat <<EOF | sudo tee /etc/modules-load.d/k8s.conf
overlay
br_netfilter
EOF
```

```

sudo modprobe overlay
sudo modprobe br_netfilter
cat <<EOF | sudo tee /etc/sysctl.d/k8s.conf
net.bridge.bridge-nf-call-iptables = 1
net.bridge.bridge-nf-call-ip6tables = 1
net.ipv4.ip_forward = 1
EOF
sudo sysctl --system

```

```

navigoinfo_id@control-plane:~$ sudo nano /etc/iscab
navigoinfo_id@control-plane:~$ cat <<EOF | sudo tee /etc/modules-load.d/k8s.conf
> overlay
br_netfilter
EOF
sudo modprobe overlay
sudo modprobe br_netfilter
cat <<EOF | sudo tee /etc/sysctl.d/k8s.conf
net.bridge.bridge-nf-call-iptables = 1
net.bridge.bridge-nf-call-ip6tables = 1
net.ipv4.ip_forward = 1
EOF
sudo sysctl --system
overlay
br_netfilter
net.bridge.bridge-nf-call-iptables = 1
net.bridge.bridge-nf-call-ip6tables = 1
net.ipv4.ip_forward = 1
* Applying /etc/sysctl.d/10-console-messages.conf ...
kernel.printk = 4 4 1 7
* Applying /etc/sysctl.d/10-ipv6-privacy.conf ...
net.ipv6.conf.all.use_tempaddr = 2
net.ipv6.conf.default.use_tempaddr = 2
* Applying /etc/sysctl.d/10-kernel-hardening.conf ...
kernel.kptr_restrict = 1
* Applying /etc/sysctl.d/10-magic-sysrq.conf ...
kernel.sysrq = 176
* Applying /etc/sysctl.d/10-network-security.conf ...
net.ipv4.conf.default.rp_filter = 2
net.ipv4.conf.all.rp_filter = 2
* Applying /etc/sysctl.d/10-pttrace.conf ...
kernel.yama.pttrace_scope = 1
* Applying /etc/sysctl.d/10-zero-page.conf ...
vm.mmap_min_addr = 65536
* Applying /usr/lib/sysctl.d/50-default.conf ...
kernel.core_uses_pid = 1
net.ipv4.conf.default.rp_filter = 2
net.ipv4.conf.default.accept_source_route = 0
sysctl: setting key "net.ipv4.conf.all.accept_source_route": Invalid argument
net.ipv4.conf.default.promote_secondaries = 1
sysctl: setting key "net.ipv4.conf.all.promote_secondaries": Invalid argument
net.ipv4.ping_group_range = 0 2147483647
net.core.default_qdisc = fq_codel
fs.protected_hardlinks = 1
fs.protected_symlinks = 1
fs.protected_regular = 1
fs.protected_fifos = 1
* Applying /usr/lib/sysctl.d/50-pid-max.conf ...
kernel.pid_max = 4194304
* Applying /etc/sysctl.d/60-gce-network-security.conf ...
net.ipv4.tcp_syncookies = 1
net.ipv4.conf.all.accept_source_route = 0
net.ipv4.conf.default.accept_source_route = 0
net.ipv4.conf.all.accept_redirects = 0
net.ipv4.conf.default.accept_redirects = 0
net.ipv4.conf.all.secure_redirects = 1
net.ipv4.conf.default.secure_redirects = 1
net.ipv4.ip_forward = 0

```

ssh.cloud.google.com/v2/ssh/projects/klaussa-5-487012/zones/asia-southeast2-a/instances/v

ssh.cloud.google.com/v2/ssh/projects/klaussa-5-487012/zones/asia-southeast



SSH-in-browser

```
No VM guests are running outdated hypervisor (qemu) binaries on this host.
navigoinfo_id@worker-1:~$ sudo nano /etc/fstab
navigoinfo_id@worker-1:~$ cat <<EOF | sudo tee /etc/modules-load.d/k8s.conf
overlay
br_netfilter
EOF
sudo modprobe overlay
sudo modprobe br_netfilter
cat <<EOF | sudo tee /etc/sysctl.d/k8s.conf
net.bridge.bridge-nf-call-iptables = 1
net.bridge.bridge-nf-call-ip6tables = 1
net.ipv4.ip_forward = 1
EOF
sudo sysctl --system
overlay
br_netfilter
net.bridge.bridge-nf-call-iptables = 1
net.bridge.bridge-nf-call-ip6tables = 1
net.ipv4.ip_forward = 1
* Applying /etc/sysctl.d/10-console-messages.conf ...
kernel.printk = 4 4 1 7
* Applying /etc/sysctl.d/10-ipv6-privacy.conf ...
net.ipv6.conf.all.use_tempaddr = 2
net.ipv6.conf.default.use_tempaddr = 2
* Applying /etc/sysctl.d/10-kernel-hardening.conf ...
kernel.kptr_restrict = 1
* Applying /etc/sysctl.d/10-magic-sysrq.conf ...
kernel.sysrq = 176
* Applying /etc/sysctl.d/10-network-security.conf ...
net.ipv4.conf.default.rp_filter = 2
net.ipv4.conf.all.rp_filter = 2
* Applying /etc/sysctl.d/10-pttrace.conf ...
kernel.yama.pttrace_scope = 1
* Applying /etc/sysctl.d/10-zero-page.conf ...
vm.mmap_min_addr = 65536
* Applying /usr/lib/sysctl.d/50-default.conf ...
kernel.core_uses_pid = 1
net.ipv4.conf.default.rp_filter = 2
net.ipv4.conf.default.accept_source_route = 0
sysctl: setting key "net.ipv4.conf.all.accept_source_route": Invalid argument
net.ipv4.conf.default.promote_secondaries = 1
sysctl: setting key "net.ipv4.conf.all.promote_secondaries": Invalid argument
net.ipv4.ping_group_range = 0 2147483647
net.core.default_qdisc = fq_codel
fs.protected_hardlinks = 1
fs.protected_symlinks = 1
fs.protected_regular = 1
fs.protected_fifos = 1
* Applying /usr/lib/sysctl.d/50-pid-max.conf ...
kernel.pid_max = 4194304
* Applying /etc/sysctl.d/60-gce-network-security.conf ...
net.ipv4.tcp_syncookies = 1
net.ipv4.conf.all.accept_source_route = 0
net.ipv4.conf.default.accept_source_route = 0
net.ipv4.conf.all.accept_redirects = 0
net.ipv4.conf.default.accept_redirects = 0
net.ipv4.conf.all.secure_redirects = 1
net.ipv4.conf.default.secure_redirects = 1
net.ipv4.ip_forward = 0
```

ssh.cloud.google.com/v2/ssh/projects/klaussa-5-487012/zones/asia-southeast2-a/instances/worker-2

SSH-in-browser

```
navigoinfo_id@worker-2:~$ sudo nano /etc/fstab
navigoinfo_id@worker-2:~$ cat <<EOF | sudo tee /etc/modules-load.d/k8s.conf
overlay
br_netfilter
EOF
sudo modprobe overlay
sudo modprobe br_netfilter
cat <<EOF | sudo tee /etc/sysctl.d/k8s.conf
net.bridge.bridge-nf-call-iptables = 1
net.bridge.bridge-nf-call-ip6tables = 1
net.ipv4.ip_forward = 1
EOF
sudo sysctl --system
overlay
br_netfilter
net.bridge.bridge-nf-call-iptables = 1
net.bridge.bridge-nf-call-ip6tables = 1
net.ipv4.ip_forward = 1
* Applying /etc/sysctl.d/10-console-messages.conf ...
kernel.printk = 4 4 1 7
* Applying /etc/sysctl.d/10-ipv6-privacy.conf ...
net.ipv6.conf.all.use_tempaddr = 2
net.ipv6.conf.default.use_tempaddr = 2
* Applying /etc/sysctl.d/10-kernel-hardening.conf ...
kernel.kptr_restrict = 1
* Applying /etc/sysctl.d/10-magic-sysrq.conf ...
kernel.sysrq = 176
* Applying /etc/sysctl.d/10-network-security.conf ...
net.ipv4.conf.default.rp_filter = 2
net.ipv4.conf.all.rp_filter = 2
* Applying /etc/sysctl.d/10-pttrace.conf ...
kernel.yama.pttrace_scope = 1
* Applying /etc/sysctl.d/10-zero-page.conf ...
vm.mmap_min_addr = 65536
* Applying /usr/lib/sysctl.d/50-default.conf ...
kernel.core_uses_pid = 1
net.ipv4.conf.default.rp_filter = 2
net.ipv4.conf.default.accept_source_route = 0
sysctl: setting key "net.ipv4.conf.all.accept_source_route": Invalid argument
net.ipv4.conf.default.promote_secondaries = 1
sysctl: setting key "net.ipv4.conf.all.promote_secondaries": Invalid argument
net.ipv4.ping_group_range = 0 2147483647
net.core.default_qdisc = fq_codel
fs.protected_hardlinks = 1
fs.protected_symlinks = 1
fs.protected_regular = 1
fs.protected_fifos = 1
* Applying /usr/lib/sysctl.d/50-pid-max.conf ...
kernel.pid_max = 4194304
* Applying /etc/sysctl.d/60-gce-network-security.conf ...
net.ipv4.tcp_syncookies = 1
net.ipv4.conf.all.accept_source_route = 0
net.ipv4.conf.default.accept_source_route = 0
net.ipv4.conf.all.accept_redirects = 0
net.ipv4.conf.default.accept_redirects = 0
net.ipv4.conf.all.secure_redirects = 1
net.ipv4.conf.default.secure_redirects = 1
net.ipv4.ip_forward = 0
net.ipv4.conf.all.send_redirects = 0
```

sudo apt-get update

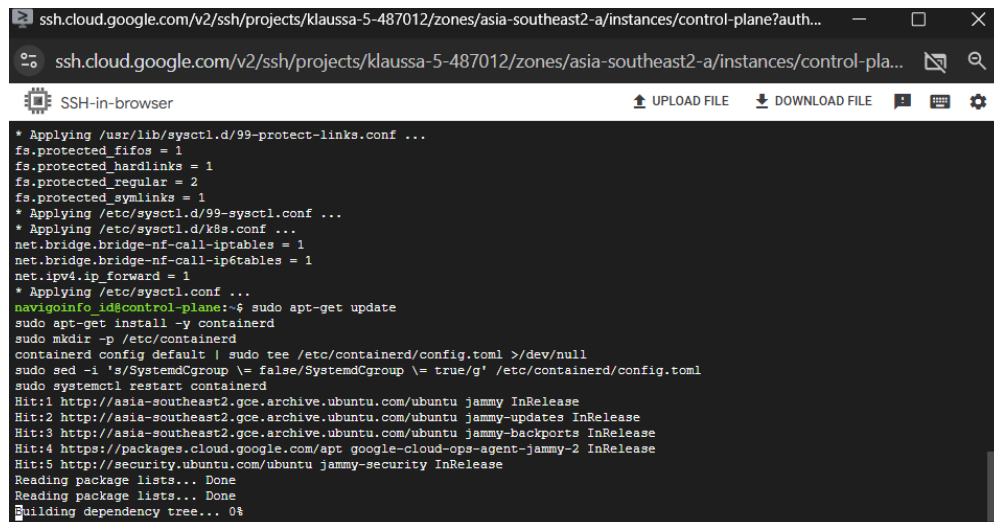
sudo apt-get install -y containerd

sudo mkdir -p /etc/containerd

containerd config default | sudo tee /etc/containerd/config.toml >/dev/null

```
sudo sed -i 's/SystemdCgroup \= false/SystemdCgroup \= true/g'
/etc/containerd/config.toml
```

```
sudo systemctl restart containerd
```



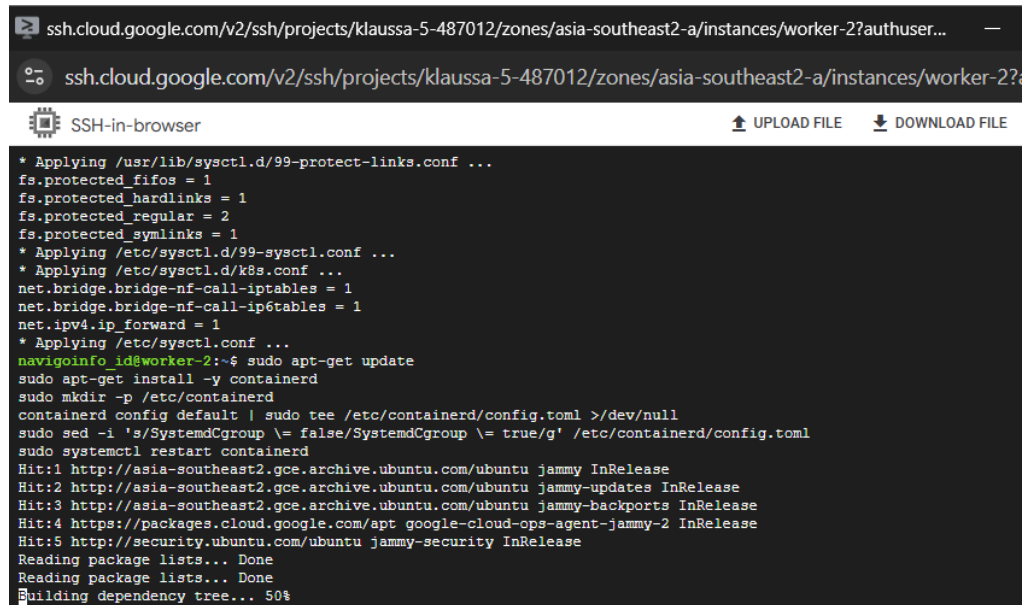
The screenshot shows a terminal window titled "SSH-in-browser" with a URL bar indicating the connection to a Google Cloud instance. The terminal output shows the following commands and their results:

```
* Applying /usr/lib/sysctl.d/99-protect-links.conf ...
fs.protected_fifos = 1
fs.protected_hardlinks = 1
fs.protected_regular = 2
fs.protected_symlinks = 1
* Applying /etc/sysctl.d/99-sysctl.conf ...
* Applying /etc/sysctl.d/k8s.conf ...
net.bridge.bridge-nf-call-iptables = 1
net.bridge.bridge-nf-call-ip6tables = 1
net.ipv4.ip_forward = 1
* Applying /etc/sysctl.conf ...
navigoinfo id@control-plane:~$ sudo apt-get update
sudo apt-get install -y containerd
sudo mkdir -p /etc/containerd
containerd config default | sudo tee /etc/containerd/config.toml >/dev/null
sudo sed -i 's/SystemdCgroup \= false/SystemdCgroup \= true/g' /etc/containerd/config.toml
sudo systemctl restart containerd
Hit:1 http://asia-southeast2.gce.archive.ubuntu.com/ubuntu jammy InRelease
Hit:2 http://asia-southeast2.gce.archive.ubuntu.com/ubuntu jammy-updates InRelease
Hit:3 http://asia-southeast2.gce.archive.ubuntu.com/ubuntu jammy-backports InRelease
Hit:4 https://packages.cloud.google.com/apt google-cloud-ops-agent-jammy-2 InRelease
Hit:5 http://security.ubuntu.com/ubuntu jammy-security InRelease
Reading package lists... Done
Reading package lists... Done
Building dependency tree... 0%
```

```
ssh.cloud.google.com/v2/ssh/projects/klaussa-5-487012/zones/asia-southeast2-a/instances/worker-1?authuser...
ssh.cloud.google.com/v2/ssh/projects/klaussa-5-487012/zones/asia-southeast2-a/instances/worker-1?a...

SSH-in-browser  UPLOAD FILE  DOWNLOAD FILE

* Applying /usr/lib/sysctl.d/50-pid-max.conf ...
kernel.pid_max = 4194304
* Applying /etc/sysctl.d/60-gce-network-security.conf ...
net.ipv4.tcp_syncookies = 1
net.ipv4.conf.all.accept_source_route = 0
net.ipv4.conf.default.accept_source_route = 0
net.ipv4.conf.all.accept_redirects = 0
net.ipv4.conf.default.accept_redirects = 0
net.ipv4.conf.all.secure_redirects = 1
net.ipv4.conf.default.secure_redirects = 1
net.ipv4.ip_forward = 0
net.ipv4.conf.all.send_redirects = 0
net.ipv4.conf.default.send_redirects = 0
net.ipv4.conf.all.rp_filter = 1
net.ipv4.conf.default.rp_filter = 1
net.ipv4.icmp_echo_ignore_broadcasts = 1
net.ipv4.icmp_ignore_bogus_error_responses = 1
net.ipv4.conf.all.log_martians = 1
net.ipv4.conf.default.log_martians = 1
kernel.randomize_va_space = 2
kernel.panic = 10
* Applying /etc/sysctl.d/99-cloudimg-ipv6.conf ...
net.ipv6.conf.all.use_tempaddr = 0
net.ipv6.conf.default.use_tempaddr = 0
* Applying /etc/sysctl.d/99-gce-strict-reverse-path-filtering.conf ...
* Applying /usr/lib/sysctl.d/99-protect-links.conf ...
fs.protected_fifos = 1
fs.protected_hardlinks = 1
fs.protected_regular = 2
fs.protected_symlinks = 1
* Applying /etc/sysctl.d/99-sysctl.conf ...
* Applying /etc/sysctl.d/k8s.conf ...
net.bridge.bridge-nf-call-iptables = 1
net.bridge.bridge-nf-call-ip6tables = 1
net.ipv4.ip_forward = 1
* Applying /etc/sysctl.conf ...
navigoinfo_id@worker-1:~$ sudo apt-get update
sudo apt-get install -y containerd
sudo mkdir -p /etc/containerd
containerd config default | sudo tee /etc/containerd/config.toml >/dev/null
sudo sed -i 's/SystemdCgroup \= false/SystemdCgroup \= true/g' /etc/containerd/config.toml
sudo systemctl restart containerd
Hit:1 http://asia-southeast2.gce.archive.ubuntu.com/ubuntu jammy InRelease
Hit:2 http://asia-southeast2.gce.archive.ubuntu.com/ubuntu jammy-updates InRelease
Hit:3 http://asia-southeast2.gce.archive.ubuntu.com/ubuntu jammy-backports InRelease
Hit:4 https://packages.cloud.google.com/apt google-cloud-ops-agent-jammy-2 InRelease
Hit:5 http://security.ubuntu.com/ubuntu jammy-security InRelease
Reading package lists... Done
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  runc
The following NEW packages will be installed:
  containerd runc
0 upgraded, 2 newly installed, 0 to remove and 6 not upgraded.
Need to get 47.4 MB of archives.
After this operation, 178 MB of additional disk space will be used.
0% [Working]
```



The screenshot shows a terminal window titled "SSH-in-browser" with a URL bar indicating a connection to a Google Cloud instance. The terminal output shows the application of various system configurations, the installation of containerd, and the execution of apt-get update and apt-get install -y containerd. It also shows the configuration of containerd and the installation of various packages including jammy, jammy-updates, jammy-backports, google-cloud-ops-agent-jammy-2, and jammy-security. The terminal output is as follows:

```
* Applying /usr/lib/sysctl.d/99-protect-links.conf ...
fs.protected_fifos = 1
fs.protected_hardlinks = 1
fs.protected_regular = 2
fs.protected_symlinks = 1
* Applying /etc/sysctl.d/99-sysctl.conf ...
* Applying /etc/sysctl.d/k8s.conf ...
net.bridge.bridge-nf-call-iptables = 1
net.bridge.bridge-nf-call-ip6tables = 1
net.ipv4.ip_forward = 1
* Applying /etc/sysctl.conf ...
navigoinfo_id@worker-2:~$ sudo apt-get update
sudo apt-get install -y containerd
sudo mkdir -p /etc/containerd
containerd config default | sudo tee /etc/containerd/config.toml >/dev/null
sudo sed -i 's/SystemdCgroup \= false/SystemdCgroup \= true/g' /etc/containerd/config.toml
sudo systemctl restart containerd
Hit:1 http://asia-southeast2.gce.archive.ubuntu.com/ubuntu jammy InRelease
Hit:2 http://asia-southeast2.gce.archive.ubuntu.com/ubuntu jammy-updates InRelease
Hit:3 http://asia-southeast2.gce.archive.ubuntu.com/ubuntu jammy-backports InRelease
Hit:4 https://packages.cloud.google.com/apt google-cloud-ops-agent-jammy-2 InRelease
Hit:5 http://security.ubuntu.com/ubuntu jammy-security InRelease
Reading package lists... Done
Reading package lists... Done
Building dependency tree... 50%
```

sudo apt-get update

sudo apt-get install -y apt-transport-https ca-certificates curl gpg

**curl -fsSL https://pkgs.k8s.io/core:/stable:/v1.31/deb/Release.key | sudo gpg
--dearmor -o /etc/apt/keyrings/kubernetes-apt-keyring.gpg**

**echo 'deb [signed-by=/etc/apt/keyrings/kubernetes-apt-keyring.gpg]
https://pkgs.k8s.io/core:/stable:/v1.31/deb/ /' | sudo tee
/etc/apt/sources.list.d/kubernetes.list**

sudo apt-get update

sudo apt-get install -y kubelet kubeadm kubectl

sudo apt-mark hold kubelet kubeadm kubectl

sudo apt-get install -y conntrack

```
ssh.cloud.google.com/v2/ssh/projects/klaussa-5-487012/zones/asia-southeast2-a/instances/control-plane?auth...
SSH-in-browser
Running kernel seems to be up-to-date.

No services need to be restarted.

No containers need to be restarted.

No user sessions are running outdated binaries.

No VM guests are running outdated hypervisor (qemu) binaries on this host.
navigoinfo id@control-plane:~$ sudo apt-get update
sudo apt-get install -y apt-transport-https ca-certificates curl gpg
curl -fsSL https://pkgs.k8s.io/core:stable/v1.31/deb/Release.key | sudo gpg --dearmor -o /etc/apt/keyrings/kubernetes-apt-keyring.gpg
echo 'deb [signed-by=/etc/apt/keyrings/kubernetes-apt-keyring.gpg] https://pkgs.k8s.io/core:stable/v1.31/deb/ /' | sudo tee /etc/apt/sources.list.d/kubernetes.list
sudo apt-get update
sudo apt-get install -y kubelet kubeadm kubectl
sudo apt-mark hold kubelet kubeadm kubectl
sudo apt-get install -y conntrack
Hit:1 http://asia-southeast2.gce.archive.ubuntu.com/ubuntu jammy InRelease
Hit:2 http://asia-southeast2.gce.archive.ubuntu.com/ubuntu jammy-updates InRelease
Hit:3 http://asia-southeast2.gce.archive.ubuntu.com/ubuntu jammy-backports InRelease
Hit:4 https://packages.cloud.google.com/apt google-cloud-ops-agent-jammy-2 InRelease
Hit:5 http://security.ubuntu.com/ubuntu jammy-security InRelease
Reading package lists... 25%
```

```
navigoinfo id@worker-1:~$ sudo apt-get update
sudo apt-get install -y apt-transport-https ca-certificates curl gpg
curl -fsSL https://pkgs.k8s.io/core:stable/v1.31/deb/Release.key | sudo gpg --dearmor -o /etc/apt/keyrings/kubernetes-apt-keyring.gpg
echo 'deb [signed-by=/etc/apt/keyrings/kubernetes-apt-keyring.gpg] https://pkgs.k8s.io/core:stable/v1.31/deb/ /' | sudo tee /etc/apt/sources.list.d/kubernetes.list
sudo apt-get update
sudo apt-get install -y kubelet kubeadm kubectl
sudo apt-mark hold kubelet kubeadm kubectl
sudo apt-get install -y conntrack
Hit:1 http://asia-southeast2.gce.archive.ubuntu.com/ubuntu jammy InRelease
Hit:2 http://asia-southeast2.gce.archive.ubuntu.com/ubuntu jammy-updates InRelease
Hit:3 http://asia-southeast2.gce.archive.ubuntu.com/ubuntu jammy-backports InRelease
Hit:4 https://packages.cloud.google.com/apt google-cloud-ops-agent-jammy-2 InRelease
Hit:5 http://security.ubuntu.com/ubuntu jammy-security InRelease
Reading package lists... 35%
```

```
ssh.cloud.google.com/v2/ssh/projects/klaussa-5-487012/zones/asia-southeast2-a/instances/worker-2?authuser...
SSH-in-browser
Running kernel seems to be up-to-date.

No services need to be restarted.

No containers need to be restarted.

No user sessions are running outdated binaries.

No VM guests are running outdated hypervisor (qemu) binaries on this host.
navigoinfo id@worker-2:~$ sudo apt-get update
sudo apt-get install -y apt-transport-https ca-certificates curl gpg
curl -fsSL https://pkgs.k8s.io/core:stable/v1.31/deb/Release.key | sudo gpg --dearmor -o /etc/apt/keyrings/kubernetes-apt-keyring.gpg
echo 'deb [signed-by=/etc/apt/keyrings/kubernetes-apt-keyring.gpg] https://pkgs.k8s.io/core:stable/v1.31/deb/ /' | sudo tee /etc/apt/sources.list.d/kubernetes.list
sudo apt-get update
sudo apt-get install -y kubelet kubeadm kubectl
sudo apt-mark hold kubelet kubeadm kubectl
sudo apt-get install -y conntrack
Hit:1 http://asia-southeast2.gce.archive.ubuntu.com/ubuntu jammy InRelease
Hit:2 http://asia-southeast2.gce.archive.ubuntu.com/ubuntu jammy-updates InRelease
Hit:3 http://asia-southeast2.gce.archive.ubuntu.com/ubuntu jammy-backports InRelease
Hit:4 https://packages.cloud.google.com/apt google-cloud-ops-agent-jammy-2 InRelease
Hit:5 http://security.ubuntu.com/ubuntu jammy-security InRelease
Reading package lists... 35%
```

sudo kubeadm init --pod-network-cidr=10.244.0.0/16

```
ssh.cloud.google.com/v2/ssh/projects/klaussa-5-487012/zones/asia-southeast2-a/instances/control-plane?auth...
ssh.cloud.google.com/v2/ssh/projects/klaussa-5-487012/zones/asia-southeast2-a/instances/control-pla...
SSH-in-browser
[UPLOAD FILE] [DOWNLOAD FILE] [RETRY] [CLOSE] [SETTINGS]

debconf: (No usable dialog-like program is installed, so the dialog based frontend cannot be used. at /usr/share/perl5/Debconf/
FrontEnd/Dialog.pm line 78.)
debconf: falling back to frontend: Readline
Scanning processes...
Scanning linux images...

Running kernel seems to be up-to-date.

No services need to be restarted.

No containers need to be restarted.

No user sessions are running outdated binaries.

No VM guests are running outdated hypervisor (qemu) binaries on this host.
navigo@control-plane:~$ sudo kubeadm init --pod-network-cidr=10.244.0.0/16
I0317 12:15:08.343036 4937 version.go:261] remote version is much newer: v1.35.2; falling back to: stable-1.31
[init] Using Kubernetes version: v1.31.14
[preflight] Running pre-flight checks
[preflight] Pulling images required for setting up a Kubernetes cluster
[preflight] This might take a minute or two, depending on the speed of your internet connection
[preflight] You can also perform this action beforehand using 'kubeadm config images pull'
W0317 12:15:08.541875 4937 checks.go:843] detected that the sandbox image "registry.k8s.io/pause:3.8" of the container runti
me is inconsistent with that used by kubeadm. It is recommended to use "registry.k8s.io/pause:3.10" as the CRI sandbox image.
```

```
No containers need to be restarted.

No user sessions are running outdated binaries.

No VM guests are running outdated hypervisor (qemu) binaries on this host.
navigo@worker-1:~$ sudo kubeadm init --pod-network-cidr=10.244.0.0/16
I0317 12:16:02.969532 4920 version.go:261] remote version is much newer: v1.35.2; falling back to: stable-1.31
[init] Using Kubernetes version: v1.31.14
[preflight] Running pre-flight checks
[preflight] Pulling images required for setting up a Kubernetes cluster
[preflight] This might take a minute or two, depending on the speed of your internet connection
[preflight] You can also perform this action beforehand using 'kubeadm config images pull'
W0317 12:16:03.254454 4920 checks.go:843] detected that the sandbox image "registry.k8s.io/pause:3.8" of the container runti
me is inconsistent with that used by kubeadm. It is recommended to use "registry.k8s.io/pause:3.10" as the CRI sandbox image.
```

```
ssh.cloud.google.com/v2/ssh/projects/klaussa-5-487012/zones/asia-southeast2-a/instances/worker-2?authuser...
ssh.cloud.google.com/v2/ssh/projects/klaussa-5-487012/zones/asia-southeast2-a/instances/worker-2?a...
SSH-in-browser
[UPLOAD FILE] [DOWNLOAD FILE] [RETRY] [CLOSE] [SETTINGS]

debconf: (No usable dialog-like program is installed, so the dialog based frontend cannot be used. at /usr/share/perl5/Debconf/
FrontEnd/Dialog.pm line 78.)
debconf: falling back to frontend: Readline
Scanning processes...
Scanning linux images...

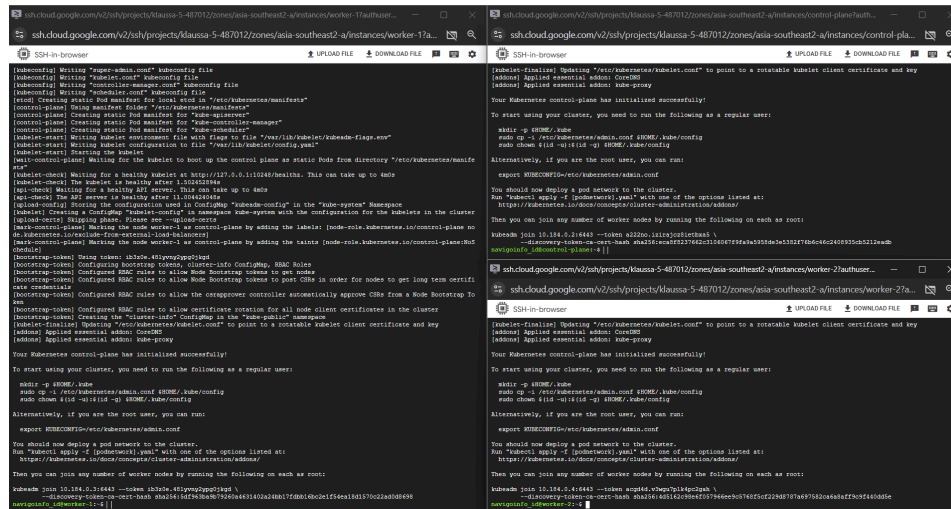
Running kernel seems to be up-to-date.

No services need to be restarted.

No containers need to be restarted.

No user sessions are running outdated binaries.

No VM guests are running outdated hypervisor (qemu) binaries on this host.
navigo@worker-2:~$ sudo kubeadm init --pod-network-cidr=10.244.0.0/16
I0317 12:15:46.299564 4942 version.go:261] remote version is much newer: v1.35.2; falling back to: stable-1.31
[init] Using Kubernetes version: v1.31.14
[preflight] Running pre-flight checks
[preflight] Pulling images required for setting up a Kubernetes cluster
[preflight] This might take a minute or two, depending on the speed of your internet connection
[preflight] You can also perform this action beforehand using 'kubeadm config images pull'
W0317 12:15:46.489403 4942 checks.go:843] detected that the sandbox image "registry.k8s.io/pause:3.8" of the container runti
me is inconsistent with that used by kubeadm. It is recommended to use "registry.k8s.io/pause:3.10" as the CRI sandbox image.
```



```
mkdir -p $HOME/.kube
sudo cp -i /etc/kubernetes/admin.conf $HOME/.kube/config
sudo chown $(id -u):$(id -g) $HOME/.kube/config
```

```
kubeadm join 10.184.0.2:6443 --token a222no.izirajcz8ietbxa5 \
--discovery-token-ca-cert-hash sha256:eca8f8237662c3106067f9fa9a5958de3e5382f76b6c46c2408935cb5212eadb
navigoinfo_id@control-plane:~$ mkdir -p $HOME/.kube
sudo cp -i /etc/kubernetes/admin.conf $HOME/.kube/config
sudo chown $(id -u):$(id -g) $HOME/.kube/config
navigoinfo_id@control-plane:~$
```

Terakhir, jalankan command berikut pada **kedua worker node** untuk memasukkan worker ke dalam cluster.

```
sudo kubeadm join 10.184.0.2:6443 --token a222no.izirajcz8ietbxa5
--discovery-token-ca-cert-hash
sha256:eca8f8237662c3106067f9fa9a5958de3e5382f76b6c46c2408935cb5212eadb
```

```

sudo rm -rf /var/lib/kubelet/
sudo rm -rf /var/lib/etcd/
navigoinfo_id@worker-1:~$ sudo kubeadm join 10.184.0.216443 --token a222no.i1zrjocsietha5 --discovery-token-ca-cert-hash sha256:a222no.i1zrjocsietha5
[preflight] Running pre-flight checks
[preflight] PFI: You can look at this config file with 'kubectl -n kube-system get cm kubeadm-config -o yaml'
[kubeadm-start] Writing kubeadm configuration to file '/var/lib/kubelet/config.yaml'
[kubeadm-start] Writing kubeadm environment file with flags to file '/var/lib/kubelet/kubeadm-flags.env'
[kubeadm-start] Starting the kubeadm
[kubeadm-check] Waiting for a healthy kubeadm at http://127.0.0.1:10248/healthz. This can take up to 4m0s
[kubeadm-check] The kubeadm is healthy after 502.783969ms
[kubeadm-start] Waiting for the kubeadm to perform the TLS Bootstrap

This node has joined the cluster:
* Certificate signing request was sent to apiclient and a response was received.
* The kubeadm was informed of the new secure connection details.

Run 'kubectl get nodes' on the control-plane to see this node join the cluster.
navigoinfo_id@worker-1:~$

```

kubectl apply -f <https://github.com/flannel-io/flannel/releases/latest/download/kube-flannel.yml>

```

navigoinfo_id@control-plane:~$ kubectl apply -f https://github.com/flannel-io/flannel/releases/latest/download/kube-flannel.yml
namespace/kube-flannel created
serviceaccount/flannel created
clusterrole.rbac.authorization.k8s.io/flannel created
clusterrolebinding.rbac.authorization.k8s.io/flannel created
configmap/kube-flannel-cfg created
daemonset.apps/kube-flannel-ds created
navigoinfo_id@control-plane:~$

```

kubectl get nodes

```

clusterrole.rbac.authorization.k8s.io/flannel created
clusterrolebinding.rbac.authorization.k8s.io/flannel created
configmap/kube-flannel-cfg created
daemonset.apps/kube-flannel-ds created
navigoinfo_id@control-plane:~$ ^C
navigoinfo_id@control-plane:~$ kubectl get nodes
NAME             STATUS    ROLES    AGE   VERSION
control-plane    Ready     control-plane   18m   v1.31.14
worker-1         Ready     <none>       5m37s v1.31.14
worker-2         Ready     <none>       5m6s  v1.31.14
navigoinfo_id@control-plane:~$

```

kubectl create deployment nginx-test --image=nginx

```

worker-2         Ready     <none>       5m0s   v1.31.14
navigoinfo_id@control-plane:~$ kubectl create deployment nginx-test --image=nginx
deployment.apps/nginx-test created
navigoinfo_id@control-plane:~$

```

kubectl expose deployment nginx-test --port=80 --type=LoadBalancer

```

navigoinfo_id@control-plane:~$ kubectl expose deployment nginx-test --port=80 --type=LoadBalancer
service/nginx-test exposed
navigoinfo_id@control-plane:~$

```

kubectl get pods -o wide

```

worker-2         Ready     <none>       5m0s   v1.31.14
navigoinfo_id@control-plane:~$ kubectl create deployment nginx-test --image=nginx
deployment.apps/nginx-test created
navigoinfo_id@control-plane:~$ kubectl expose deployment nginx-test --port=80 --type=LoadBalancer
service/nginx-test exposed
navigoinfo_id@control-plane:~$ kubectl get pods -o wide
NAME             READY   STATUS    RESTARTS   AGE   IP           NODE       NOMINATED NODE   READINESS GATES
nginx-test-sb77bfd686-c8rj9  1/1     Running   0          62s   10.244.2.2   worker-2   <none>           <none>
navigoinfo_id@control-plane:~$

```

2. Jelaskan apa itu Horizontal Pod Autoscaler dan perannya dalam melakukan scaling pada aplikasi.

Horizontal Pod Autoscaler adalah fitur Kubernetes yang secara otomatis menambah atau mengurangi jumlah replika pod berdasarkan metrik tertentu, paling umum adalah CPU usage atau memory usage.

Cara kerjanya:

- HPA secara periodik (default tiap 15 detik) mengambil metrik dari Metrics Server
- Jika rata-rata CPU usage pod melebihi threshold yang ditentukan (misal 70%), HPA akan scale out — menambah jumlah pod
- Jika beban berkurang, HPA akan scale in — mengurangi pod kembali

3. Misalnya Anda adalah pemilik suatu data center yang memiliki banyak VM dengan beragam spesifikasi.

a. Hal-hal apa saja yang akan Anda jadikan pertimbangan dalam menentukan server mana yang menjadi control plane dan mana yang menjadi worker?

Dalam memilih VM mana yang menjadi control plane dan mana yang menjadi worker, ada beberapa hal yang perlu dipertimbangkan.

Dari sisi CPU, control plane tidak memerlukan CPU yang terlalu tinggi, namun harus stabil, sedangkan worker node membutuhkan CPU yang lebih tinggi karena tugasnya menjalankan container. Dari sisi RAM, control plane cukup dengan sekitar 4–8 GB, sementara worker node membutuhkan RAM lebih besar sesuai beban aplikasi yang dijalankan. Untuk storage, control plane sebaiknya menggunakan SSD yang cepat karena etcd sangat sensitif terhadap I/O, sedangkan worker node cukup disesuaikan dengan kebutuhan image dan volume. Dari sisi ketersediaan, control plane harus highly available dan tidak boleh dijadikan VM yang sering restart, sedangkan worker node lebih fleksibel. Jumlah control plane sebaiknya ganjil (1, 3, atau 5) untuk memenuhi quorum etcd, sedangkan jumlah worker disesuaikan dengan kebutuhan beban kerja. Terakhir dari sisi keamanan, control plane harus diisolasi dari jaringan publik, sementara worker node bisa lebih terbuka sesuai kebutuhan service.

Intinya, control plane butuh stabilitas dan storage cepat, sementara worker butuh compute power yang tinggi.

- b. Step-step mana saja yang mungkin dapat Anda *skip* bila membuat cluster melalui *cloud provider* yang sudah memiliki *dedicated service*? Berikan penjelasannya.

Beberapa langkah dalam tutorial ini tidak perlu dilakukan jika menggunakan managed Kubernetes dari cloud provider.

Menonaktifkan swap tidak perlu dilakukan karena cloud provider sudah mengkonfigurasi VM sesuai kebutuhan Kubernetes. Konfigurasi kernel modules seperti overlay dan br_netfilter juga bisa dilewati karena node image yang disediakan provider sudah dikonfigurasi sejak awal. Instalasi containerd tidak diperlukan karena container runtime sudah terinstall di dalam node image. Instalasi kubeadm, kubelet, dan kubectl untuk keperluan setup cluster juga tidak perlu dilakukan karena cluster dibuat secara otomatis oleh managed service; kubectl hanya perlu diinstall di local machine untuk mengakses cluster. Perintah kubeadm init bisa dilewati karena control plane sepenuhnya dibuat dan dikelola oleh provider. Begitu pula kubeadm join tidak diperlukan karena worker node ditambahkan secara otomatis melalui node pool atau auto-scaling group. Instalasi Flannel sebagai CNI juga tidak diperlukan karena provider memiliki CNI sendiri seperti AWS VPC CNI atau GKE built-in CNI. Terakhir, instalasi MetalLB tidak dibutuhkan karena cloud provider sudah memiliki native Cloud Load Balancer seperti AWS ELB atau GCP CLB yang langsung terintegrasi.

Yang tetap perlu dilakukan adalah deploy aplikasi dan konfigurasi YAML seperti Deployment, Service, dan HPA.

- c. Pada tutorial ini, kita mengekspos aplikasi secara langsung melalui metalLB. Apabila kita meng-handle data sensitif, apa yang Anda akan lakukan untuk menambahkan *security layer* antara metalLB dan pods?

Jika kita menangani data sensitif, mengekspos pod secara langsung melalui MetalLB sangat berisiko. Ada beberapa layer keamanan yang perlu ditambahkan.

Pertama, gunakan Ingress Controller dengan TLS termination. Tempatkan Nginx Ingress Controller atau Traefik di antara MetalLB dan pod, sehingga MetalLB hanya mengekspos IP ke Ingress Controller, bukan langsung ke pod. Dengan ini

kita bisa menerapkan HTTPS/TLS untuk enkripsi in-transit dan menggunakan cert-manager untuk pengelolaan sertifikat SSL secara otomatis.

Kedua, terapkan Network Policy dari Kubernetes untuk membatasi pod mana yang boleh berkomunikasi satu sama lain, mengikuti prinsip least privilege.

Ketiga, tambahkan Web Application Firewall atau WAF, misalnya melalui plugin ModSecurity pada Nginx Ingress, untuk memfilter request berbahaya seperti SQL injection dan XSS.

Keempat, gunakan service mesh seperti Istio atau Linkerd untuk mengenkripsi komunikasi antar pod menggunakan mutual TLS (mTLS), sehingga east-west traffic di dalam cluster pun tetap aman.

Kelima, konfigurasi rate limiting di level Ingress agar pod tidak bisa dibanjiri request secara berlebihan.

Arsitektur yang direkomendasikan adalah: Internet → MetalLB (IP publik) → Ingress Controller (TLS + WAF) → Service → Pods.